

Decision Validation Before Execution: A Closed-Loop Simulate-and-Stress-Test Architecture for Water Infrastructure Control

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Abstract

Water infrastructure control commonly operates reactively: anomalies are detected, sometimes predicted, but rarely simulated and stress-tested before a response is executed. This report presents AquaGuardian, a deterministic proof-of-concept platform that inserts an explicit simulate-stress-test-validate step — conceptually related to runtime-assurance safety filtering used in autonomous and aerospace control — between decision and execution in the water domain. Across five simulated water-risk scenarios under four stress profiles, comparing reactive, detection-only, and closed-loop policies, the closed-loop policy raised the stress-test pass rate of its selected plan from 0% to 100%, reduced simulated water loss by 72–74% (depending on aggregation method) and response delay by 99.9% relative to the reactive baseline. One metric — nominal safety-risk score in a single scenario — did not improve and is reported as an open question rather than resolved. All results are software-simulated (TRL 3); no field validation has been performed. Methodology, limitations, and a validation roadmap toward field evidence are described in full below.

1. Introduction

Water infrastructure is disproportionately managed reactively: leaks are typically discovered after significant loss, pump failures after service interruption, and irrigation is frequently scheduled on fixed intervals rather than measured plant stress. This pattern has real cost — agriculture alone consumes roughly 70% of global freshwater, and global demand is projected to rise 20–30% by 2050, against a backdrop where 2.1 billion people still lack access to safe drinking water. Digital tools increasingly detect and predict water-system anomalies (Section 2.1), but a further step — simulating a candidate response and stress-testing it against adversarial conditions before it reaches physical equipment — is comparatively underexplored in the water domain relative to safety-critical autonomous-control domains where it is well established (Section 2.2). This report investigates whether that step, implemented as an explicit closed-loop cycle, measurably improves decision quality in simulation, and documents the resulting proof-of-concept honestly at its current TRL 3 maturity.

2. Related Work

2.1 Digital Twins and Machine Learning for Water Distribution Networks

A substantial and active literature applies digital twins and machine learning to leak detection and localization in water distribution networks (WDNs). Pandey et al. develop a machine learning-based digital twin using single- and two-stage anomaly-detection models (logistic regression, random forest) to detect and localize leaks from pressure-sensor data on a real campus network [1]. Singha, Singh, and Maheshwari propose an EPANET/WNTR-based digital-twin framework validated on a real campus WDN, combining hydraulic modeling with a graph neural network for leak localization, reporting approximately 90% leak-detection accuracy on their test zone [2, 3]. Related work applies genetic algorithms over calibrated EPANET models for online leak diagnosis and sizing [4], and Mücke et al. apply generative deep learning to produce probabilistic leak-location estimates under sensor noise and limited instrumentation [5]. A decision-support framing closer to AquaGuardian's own appears in

Brahmbhatt, Maheshwari, and Gudi, who couple a digital twin to a decision-support system for water-quality regulation and leak localization [6]. Across this literature, the emphasis is consistently on *detecting and localizing* the anomaly itself; this review did not find published work that additionally simulates and stress-tests a specific candidate operational response — e.g. which valve to close, or whether a closure would violate a constraint elsewhere in the network — before that response is executed.

2.2 Runtime Assurance and Safety Filtering in Autonomous Control

The idea of filtering a candidate control action against safety constraints before it reaches a physical plant is well established outside the water domain, under the name Runtime Assurance (RTA) or safety filtering. The Simplex architecture formalized pairing a high-performance controller with a verified baseline that takes over when safety is threatened [7]; RTA has since been extended to autonomous driving, spacecraft, and aircraft control, where a monitor checks a primary controller's desired action against the system state and substitutes a safe fallback if the desired action would violate a safety boundary [8, 9, 10, 11]. This body of work motivated the framing used here: AquaGuardian's Stress Test / reject / re-optimize loop is structurally the same pattern — an online check of a candidate action against constraints, with rejection and fallback when the check fails — applied to water-infrastructure actions (valve closures, pump load changes, irrigation schedules) rather than aircraft or spacecraft trajectories. To the best of the author's knowledge from this review, this specific pattern has not been previously documented as applied to water-infrastructure control; this is offered as a scoping observation from a non-exhaustive literature review conducted for this submission, not a formal claim of novelty.

3. Research Question

Does inserting an explicit simulate-stress-test-validate step before execution measurably improve decision outcomes — specifically, constraint-violation (stress-test failure) rate, avoidable water loss, and response latency — relative to a reactive baseline and a detection-only AI policy, across a simulated water-risk scenario set designed to include adversarial and degraded conditions?

4. Methodology

Simulator design. Five water-risk scenarios were implemented (pipeline leak, pump degradation, water contamination, drought stress, wildfire risk), each with 3–4 domain-relevant candidate actions (e.g. isolate valve, throttle pump, escalate irrigation, dispatch mist/drone response). Each scenario is evaluated under four stress profiles — nominal, noisy sensors, high severity, and offline/degraded connectivity — deliberately including adversarial and degraded conditions rather than only the nominal case.

Policies compared. (1) *Reactive baseline*: acts only after a threshold breach, with a modeled 300-second response delay. (2) *Detection-only AI*: anomaly detection followed by a fixed rule-based response, with no candidate-action simulation or stress testing. (3) *AquaGuardian closed loop*: evaluates multiple candidate actions in a digital twin under all four stress profiles before selecting one; a candidate that fails stress testing is rejected and the search continues.

Metrics. Risk detection score, decision latency, percentage of stress profiles passed, and scenario-specific proxy metrics: water-loss proxy, energy-consumption proxy, service disruption, contamination exposure, crop stress, and wildfire-risk proxy. All are explicitly proxy metrics computed by the simulator's own scoring functions, not measurements from physical infrastructure.

Design limitation, stated directly: each (scenario, stress-profile, policy) combination is evaluated as a single deterministic run, not a repeated stochastic trial. Consequently, the results below are point estimates without confidence intervals or significance testing — a methodological limitation discussed further in Section 7.

5. Results

5.1 Aggregate comparison across 5 scenarios × 4 stress profiles (20 runs per policy, 60 total):

Policy	Avg. water loss (L)	Avg. stress-test pass rate	Avg. safety risk score	Avg. response delay (s)
Reactive baseline	420.96	0%	0.100	300.0
Detection-only AI	194.33	50%	0.134	2.0
AquaGuardian closed-loop	117.04	100%	0.112	0.25

Water-loss improvement is reported two ways for transparency: 72.2% comparing aggregate averages (117.04 vs. 420.96 L), or 74.4% averaging the five per-scenario percentage reductions (72.9%, 99.9%, 68.4%, 79.1%, 51.9%) — both are legitimate, differently-weighted summaries of the same data.

5.2 The one metric that does not favor the closed-loop policy: in the pump-degradation scenario, the closed loop's selected plan carries a **higher** nominal safety-risk score than the reactive baseline (a −111.1% "change"), even though it is the only policy whose plan passes stress testing. This is reported as-is, not adjusted, and discussed in Section 6.

5.3 Stress-test validator coverage (independent of policy selection): across 80 (scenario, stress-profile, candidate-action) combinations covering 12 distinct candidate actions, 42 passed and 38 failed (52.5%) — indicating the validator actively rejects roughly half of all candidate actions evaluated, including nominal ones, rather than acting as a rubber stamp.

5.4 Supporting hardware/protocol-layer evidence (edge decision latency, network SLA compliance, and a live oneM2M CRUD round-trip) is reported in the companion Technical Report, Section 4, and PoC Test Report, Section 5; it is out of scope for this research-methodology report beyond noting that the reflex-kernel and MEC-cognitive-layer classes exercised there are reused unmodified between this project's water domain and a separate aviation domain in the same codebase.

6. Discussion

The results are consistent with the research question's premise across four of five scenarios and all aggregate metrics: an explicit simulate-stress-test-validate step reduced simulated water loss, cut response latency by roughly three orders of magnitude (a modeled property of the reactive baseline's 300-second delay assumption rather than a claim about real network physics), and raised the stress-test pass rate of the selected plan to 100% across every scenario tested.

The pump-degradation safety-risk result (Section 5.2) is treated here as an open question rather than a confirmed finding. One plausible interpretation is that the closed loop selects a more decisive action — trading a higher nominal risk-score under this particular scoring model for actually passing validation under adversarial load conditions the reactive plan fails. This interpretation is *not* independently confirmed; an equally plausible alternative is that the safety-risk scoring function itself weights this scenario's constraints in a way that does not track real-world risk well. Distinguishing between these requires either a redesigned scoring function or field data — neither of which this PoC has produced. It is reported rather than smoothed over because a single favorable-looking aggregate number would misrepresent what the simulator actually shows.

7. Threats to Validity

- **Construct validity:** all metrics (water-loss proxy, safety-risk score, etc.) are proxy quantities defined by the simulator's own scoring functions, not measured on physical infrastructure. They may not track the real-world quantities they are named after.
- **Single-implementer bias:** the system, the three compared policies, and the metrics were all designed and implemented by the same person. No independent party has reviewed, replicated, or adversarially tested the simulator or the comparison methodology.
- **Statistical validity:** each condition is a single deterministic run; there are no repeated stochastic trials, confidence intervals, or significance tests. The reported percentages describe this specific simulation, not a distribution of outcomes.
- **External validity:** the scenario and stress-profile space (5 scenarios $\times$ 4 profiles) is illustrative, not exhaustive or drawn from a real utility's incident history. Hydraulic, agronomic, and fire-spread models are simplified and uncalibrated against any real network.
- **Reactive-baseline framing:** the 300-second reactive delay and fixed-rule detection-only behavior are simulator assumptions chosen by the same author who built the closed-loop system being compared against them; a differently specified baseline could narrow or widen the reported gap.
- **Hardware/protocol evidence scope:** the sub-1ms and network-SLA figures referenced in Section 5.4 come from a software architecture simulator, not measurements from physical embedded hardware or a real radio link.

8. Future Work

The validation pathway from this TRL 3 result toward field evidence is: (1) *TRL 4* — connect real pressure, flow, soil-moisture, and vibration sensors; replay recorded fault traces; calibrate simulation parameters against measured behavior; verify fail-safe response under sensor noise and packet loss; (2) *TRL 5* — a shadow-mode pilot alongside an operating farm, irrigation district, or small utility, comparing recommendations to actual operator actions without issuing actuator commands, to obtain the first independent, non-simulated evidence; (3) *TRL 6* — a supervised pilot on low-risk actions with independent security and safety review. Equally important as a research matter: an independent replication of the comparison methodology by a party other than the original author, and a sensitivity analysis of the safety-risk scoring function against the anomaly reported in Section 6.

9. Conclusion

Within the bounds of a deterministic software simulation, inserting an explicit simulate-stress-test-validate step before execution — a pattern conceptually related to runtime assurance in autonomous control, applied here to water infrastructure — measurably improved constraint-violation rate, water-loss proxy, and response-latency proxy relative to reactive and detection-only baselines, in four of five tested scenarios outright and with one open, honestly disclosed exception. These results support continued investment in field validation; they do not, on their own, support any claim about real-world performance, and are not presented as such.

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